

Astro HW 12

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12.1

a)

$$L = 1.2L_{\odot} \left(\frac{M}{M_{\odot}} \right)^{3.92}$$

$$L_{edd} = \frac{4\pi GM_p c}{\sigma_e}$$

$$\sigma_e = 6.55 \times 10^{-29} m^2, \quad m_p = 1.67 \times 10^{-27} kg$$

$$M_{max} = 3.514 \times 10^1 L_{\odot}$$

12.2 Accretion disk temperatures

a)

b)

Given ...

$$L_e = 4\pi r^2 \sigma_{sb} T^4$$

$$r = \frac{6GM}{c^2}$$

$$r \propto M, \quad M \propto L_E$$

We can find ...

$$L_e = 4\pi \left(\frac{6GM}{c^2} \right)^2 \sigma_{sb} T^4$$

By proportionality ...

$$\Rightarrow M^2 T^4 \propto M$$

$$\Rightarrow T^4 \propto \frac{M}{M^2}$$

$$T = M^{-1/4}$$

c)

Given ...

$$\begin{aligned}\dot{M} &= \dot{m} \left(\frac{4\pi GM m_p}{\sigma_e \eta c} \right) \\ \dot{m} &= 1 \quad \eta \sim .1 \\ r &= \frac{(6GM)}{c^2} \\ T(r) &\approx \left[\frac{GM\dot{M}}{4\pi\sigma_{SB}r^3} \right]^{1/4}\end{aligned}$$

We find ...

$T_{4M_\odot} = 1.755 \times 10^7 \text{ K}, \quad T_{10^9 M_\odot} = 1.396 \times 10^5 \text{ K}$
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d)

We find that they lie in the microwave and radio wave spectrum.

12.3 Stellar close encounters

a)

b)

$$t = \frac{1}{n_*(\pi R_{eff}^2)}$$

c)

Divide answer of above by 5

12.4 Expansion of a Newtonian universe

a)

Given ...

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G \rho^{c,0}}{3} \frac{1}{a^3}$$

We can calculate ...

$$\begin{aligned}\dot{a}^2 &= \frac{8\pi G \rho^{c,0}}{3} \frac{1}{a} \\ \frac{8\pi G \rho^{c,0}}{3} &= k \\ \dot{a} &= \sqrt{\frac{k}{a}} \\ \frac{da}{dt} &= \sqrt{\frac{k}{a}}\end{aligned}$$

$$da = \sqrt{\frac{k}{a}} dt$$

$$\sqrt{a} da = \sqrt{k} dt$$

$$\int \sqrt{a} da = \int \sqrt{k} dt$$

$$\frac{2}{3} a^{3/2} = t \sqrt{k}$$

$$a = \left(\frac{3}{2} t \sqrt{k} \right)^{2/3}$$

$$a(t) = \left(\frac{3}{2} t \sqrt{\frac{8\pi G \rho_{c,0}}{3}} \right)^{2/3}$$

b)

c)

Given ...

$$\frac{1}{t_0} = \frac{1}{H_0}$$

We find ...

$$T_{uni} = 13.97 \text{ Gyr}$$

12.5 Olbers's paradox again

a)

Given ...

$$n_{wd} = \frac{3H_0^2}{8\pi G}$$

We find

$$n_{wd} = 6.613 \times 10^{-57} \text{ m}^3$$

b)